Functional Annotation of PP_0262 (UniProt Q88R72) in *Pseudomonas putida* KT2440

Target: PP_0262 / Q88R72 · *Pseudomonas putida* (strain ATCC 47054 / DSM 6125 / KT2440) · 151 aa, ~17 kDa **UniProt annotation (baseline):** "Thioesterase putative domain-containing protein" (HotDog superfamily; Pfam PF09500 *YiiD_C*; InterPro IPR029069 HotDog_dom_sf, IPR012660 YiiD_C; TIGR02447; eggNOG COG2050)

---

1. Summary (Answer to the Research Question)

PP_0262 is *madB*, encoding MadB — a malonyl-[acyl-carrier-protein] decarboxylase (malonyl-ACP decarboxylase). Despite a generic UniProt annotation as a "putative thioesterase," PP_0262 is a **characterized enzyme with a solved 1.04 Å crystal structure (PDB 8AYV)** and a dedicated primary study (McNaught et al., *Metabolic Engineering* 2023, P 36796578).

Its primary function is to **catalyze the decarboxylation of malonyl-ACP to acetyl-ACP + CO_2**, thereby generating the acetyl-ACP "primer" that **initiates type II (dissociated) fatty acid biosynthesis (FAS II)**. This is a **FabH-independent initiation route**: it bypasses the canonical condensation of acetyl-CoA with malonyl-ACP by β-ketoacyl-ACP synthase III (FabH). MadB is **one of three parallel FAS-initiation pathways** operating in KT2440 (alongside FabH1 and FabH2). The enzyme acts in the **cytoplasm** on an ACP-tethered substrate, adopts a **HotDog fold** (the standalone C-terminal *YiiD_C* domain), and its homologs are **conserved throughout the Bacteria/Proteobacteria**.

> ⚠ **Identity verification (as required):** The gene symbol "PP_0262" and the organism (*P. putida* KT2440) match the UniProt record exactly. The HotDog/thioesterase-fold domain (PF09500 YiiD_C) is precisely the fold used by MadB. The identity is confirmed by an unambiguous structural link: UniProt Q88R72 reference [3] cites **PDB 8AYV**, whose RCSB

> title is *"Crystal structure of the Malonyl-ACP Decarboxylase MadB from Pseudomonas putida"* (gene = PP_0262, taxid 160488). **No ambiguity: this report concerns the correct protein.**

---

2. Molecular Function — The Reaction Catalyzed

2.1 Reaction and substrate specificity

MadB catalyzes:

> **malonyl-ACP → acetyl-ACP + CO_2**

- **Substrate:** malonyl-ACP (malonate thioester on the phosphopantetheine arm of acyl carrier protein). The substrate is **ACP-tethered**, not a free CoA thioester.
- **Product:** acetyl-ACP, the C_2 starter unit for fatty-acid elongation.
- **Class:** a **decarboxylase (lyase)**, *not* a hydrolytic thioesterase — an important distinction, because the protein's structural scaffold (HotDog fold, "thioesterase/thiol-ester dehydrase-isomerase" superfamily, SUPFAM SSF54637) is shared with true thioesterases, which is why databases default to a "putative thioesterase" label.
- **Downstream fate of the product:** acetyl-ACP is condensed with a second malonyl-ACP by the elongation-type β-ketoacyl-ACP synthases (KAS I/II; FabB/FabF) to yield acetoacetyl-ACP, which then enters the standard FAS II reduction/dehydration/reduction cycle (FabG → FabA/FabZ → FabI). In *P. putida* F1, the FabB synthase itself can perform both the decarboxylation and the subsequent condensation (Guo et al. 2024, P 38335573), underscoring that the acetyl-ACP made by MadB feeds directly into elongation.

2.2 Mechanism and catalytic residues

McNaught et al. (2023) determined the mechanism by combining **exhaustive in vivo alanine-scanning mutagenesis, in vitro biochemical assays, 1.04 Å X-ray crystallography (PDB 8AYV), and QM/computational modeling**. The active site sits in the HotDog-fold cavity. Key conserved, functionally essential residues (P. putida MadB numbering) are **Asn43, Asn45,**

Phe51, and Arg124; **Asn45→Ala abolishes decarboxylase activity**. These positions are present and conserved in the Q88R72 sequence (…^{41}AANVNHKSTM^{50}, ^{51}F…, …R^{122}AR^{124}…) and are retained even in distant homologs (~34 % identity) in *Xanthomonas*, confirming their catalytic importance (Yan et al. 2025, P 40197034). Arg124 is positioned to stabilize the developing negative charge/carboxylate during decarboxylation; the asparagines and Phe51 shape the oxyanion/substrate pocket — a mechanism analogous to other HotDog-fold decarboxylases.

2.3 Biochemical assay support

MadB-family activity is measured by reconstituting FAS in vitro: with malonyl-ACP as sole acyl donor plus FabB/FabG (± FabA/FabI) and NAD(P)H, MadB drives production of long-chain acyl-ACPs and NADPH oxidation (340 nm), whereas no-MadB controls do not (Yan et al. 2025). Reported specific activities for orthologs are high (e.g., *Xanthomonas oryzae* MadB ≈ 135 $nmol \cdot \mu g^{-1} \cdot s^{-1}$). MadB-type enzymes do **not** catalyze the FabH-type condensation of acetyl-CoA with malonyl-ACP (they make no butyryl-ACP in that assay), confirming a decarboxylase — not a KAS — activity.

---

3. Biological Process / Pathway

- **Pathway:** de novo **fatty acid biosynthesis (type II FAS)** — specifically the **initiation** step.
- **Role:** provides the acetyl-ACP primer that seeds the elongation cycle, an **alternative to FabH-catalyzed initiation**. In KT2440 there are **three initiation routes**: (i) FabH1 (a canonical KAS III using short-chain acyl-CoAs/acetyl-CoA), (ii) FabH2 (a KAS III accepting short- and medium-chain acyl-CoAs), and (iii) **MadB** (malonyl-ACP decarboxylation). This redundancy explains why single deletions are tolerated (e.g., growth of KT2440 is not fully blocked by deleting *madB* together with *fabH1*).
- **Precedent and relationships:** MadB is the *P. putida* counterpart of *E. coli* **MadA** (the *yiiD* product), the first-described malonyl-ACP decarboxylase that rescues *E. coli* ΔfabH strains. MadA is a **two-domain** protein (N-terminal GNAT/N-acetyltransferase domain fused to the C-terminal HotDog *YiiD_C* decarboxylase domain), whereas **MadB is the standalone HotDog domain**, which is by itself sufficient for decarboxylase activity.

- **Downstream metabolic connections:** the acetyl-ACP/acetyl groups generated by this route also supply acetylation reactions such as **UDP-GlcNAc synthesis** (feeding cell-envelope/EPS biosynthesis) in some proteobacteria; in *Xanthomonas*, *madB* loss reduces exopolysaccharide production, biofilm, and virulence, illustrating how the primary FAS-initiation activity ramifies into envelope/lipid physiology (Yan et al. 2025). These are secondary/pleiotropic consequences of the **primary decarboxylase function**.

---

4. Subcellular Localization

MadB functions in the **bacterial cytoplasm**. Rationale/evidence: - The entire FAS II system (ACP, FabD, FabH, FabB/F, FabG, FabA/Z, FabI) is soluble and cytoplasmic, and MadB's substrate is the **cytosolic, ACP-tethered malonyl-ACP**. - The 151-aa sequence contains **no signal peptide and no transmembrane segment**; the protein crystallized as a soluble globular HotDog-fold domain (PDB 8AYV, from recombinant cytoplasmic expression). - Consistent with a soluble cytoplasmic enzyme, orthologs are purified as soluble His-tagged proteins for in vitro assays.

---

5. Structure and Evolution (Evidence from Structure/Bioinformatics)

- **Fold:** single **HotDog fold** (Gene3D 3.10.129.10 "Hotdog Thioesterase"; SUPFAM SSF54637). Domain family = **YiiD C-terminal domain** (Pfam PF09500; InterPro IPR012660; NCBIfam/TIGR TIGR02447 *yiiD_Cterm*; orthology COG2050).
- **Experimental structure: PDB 8AYV**, X-ray, **1.04 Å** (near-atomic resolution), chains A/B (residues 1–151); UniProt evidence level **PE1 (protein level)**.
- **Architecture insight:** the standalone MadB domain corresponds to the catalytic C-terminal half of the E. coli MadA (YiiD) fusion protein, showing the GNAT domain is dispensable for decarboxylase catalysis.
- **Conservation:** functional MadB/MadA homologs are **widespread across domain Bacteria** and **highly conserved within Proteobacteria**; catalytic residues (Asn43/Asn45/Phe51/Arg124) are retained even at ~30 % sequence identity, a strong evolutionary signature of a conserved catalytic mechanism.

6. Supported and Refuted Hypotheses

Hypothesis	Status	Basis
PP_0262 is merely a "putative thioesterase" of unknown function	**Refuted**	It is a characterized enzyme (MadB) with solved structure and mechanism.
PP_0262 = MadB malonyl-ACP decarboxylase	**Supported (high confidence)**	PDB 8AYV title + gene PP_0262; McNaught 2023.
Primary reaction is hydrolytic thioester cleavage (thioesterase)	**Refuted**	Enzyme is a decarboxylase (malonyl-ACP → acetyl-ACP + CO_2); it does not act as a hydrolase in assays.
Function is FAS **initiation** (provides acetyl-ACP primer)	**Supported**	In vivo complementation of ΔfabH strains; in vitro reconstitution; alanine scan.
MadB is essential/sole initiator in KT2440	**Refuted**	KT2440 has three redundant initiation routes (FabH1, FabH2, MadB).
Acts in the cytoplasm on ACP-tethered substrate	**Supported**	Soluble FAS II machinery; no signal/TM sequence; soluble recombinant protein.

7. Limitations and Future Directions

- The definitive primary paper (McNaught et al. 2023, P 36796578) is **not open-access**; catalytic-residue specifics and kinetic constants were corroborated here from the structure metadata (PDB 8AYV), the abstract, and open follow-up work (Guo 2024; Yan 2025). Exact *k*cat/*K*m for KT2440 MadB were not retrievable in this session.
- The **strict physiological indispensability** of MadB in KT2440 under different carbon sources/growth conditions warrants targeted single- and double-mutant characterization (redundancy with FabH1/FabH2 and possibly FabB complicates loss-of-function phenotypes).

- Whether KT2440 MadB, like FabB in *P. putida* F1, contributes to **branched/unsaturated** fatty acid initiation or channels acetyl-ACP into UDP-GlcNAc/envelope pathways in vivo remains to be quantified.
- The regulatory logic of *madB* expression in KT2440 (operon context, inducers) is not yet defined.

References

1. McNaught KJ, Kuatsjah E, Zahn M, *et al.* **Initiation of fatty acid biosynthesis in *Pseudomonas putida* KT2440.** *Metab Eng* 2023;76:193–203.

   P 36796578

   (Defines MadB/PP_0262; PDB 8AYV.)
2. Guo S, Zhong Q, Dong Q, Cronan JE, Wang H. **Diversity in fatty acid elongation enzymes: The FabB long-chain β-ketoacyl-ACP synthase I initiates fatty acid synthesis in *Pseudomonas putida* F1.** 2024.

   P 38335573

   (MadA/MadB context; malonyl-ACP → acetyl-ACP initiation.)
3. Yan Y, Yu, Luo, Su, Ma, Hu, Wang. **Functional disparities of malonyl-ACP decarboxylase between *Xanthomonas campestris* and *Xanthomonas oryzae*.** 2025.

   P 40197034

   (MadB conservation, catalytic residues, standalone YiiD_C domain, physiology.)
4. Nelson KE, *et al.* **Complete genome sequence… *Pseudomonas putida* KT2440.** *Environ Microbiol* 2002;4:799–808.

   P 12534463

   (Genome/locus PP_0262.)
5. RCSB PDB **8AYV** — *Crystal structure of the Malonyl-ACP Decarboxylase MadB from Pseudomonas putida* (X-ray, 1.04 Å).
6. UniProtKB **Q88R72** (Q88R72_PSEPK); Pfam PF09500 (YiiD_C); InterPro IPR012660/IPR029069; TIGR02447; eggNOG COG2050.